DI32007 – Industry and Innovation

Business Model Canvas

Group 1

SoftStop EV: Intelligent Smooth-Stop System

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Introduction and poster; Value proposition; Channel; Customer relationships; Report summarization	Technical assistance; Introduction and poster; Value proposition; Key resources; key activities; Report summarization	Value proposition; Revenue streams	Customer segments	Cost structure	Key partnerships

Content

(AI statement: AI tools were used to support language polishing and visual icon generation in this report. But all key ideas, business analysis, technical content and final decisions were developed, reviewed and approved by the group members)

1 Introduction and poster

This report presents a Business Model Canvas for SoftStop EV, an intelligent smooth-stop sensor system designed for the front brake calipers of new energy vehicles. As competition in the EV market increases, ride comfort and intelligent chassis control have become important ways for manufacturers to differentiate their products. SoftStop EV addresses the low-speed braking "nodding" problem by combining speed sensing, acceleration monitoring, temperature compensation and ECU-based control logic to reduce braking jerk and improve passenger comfort [1][2].

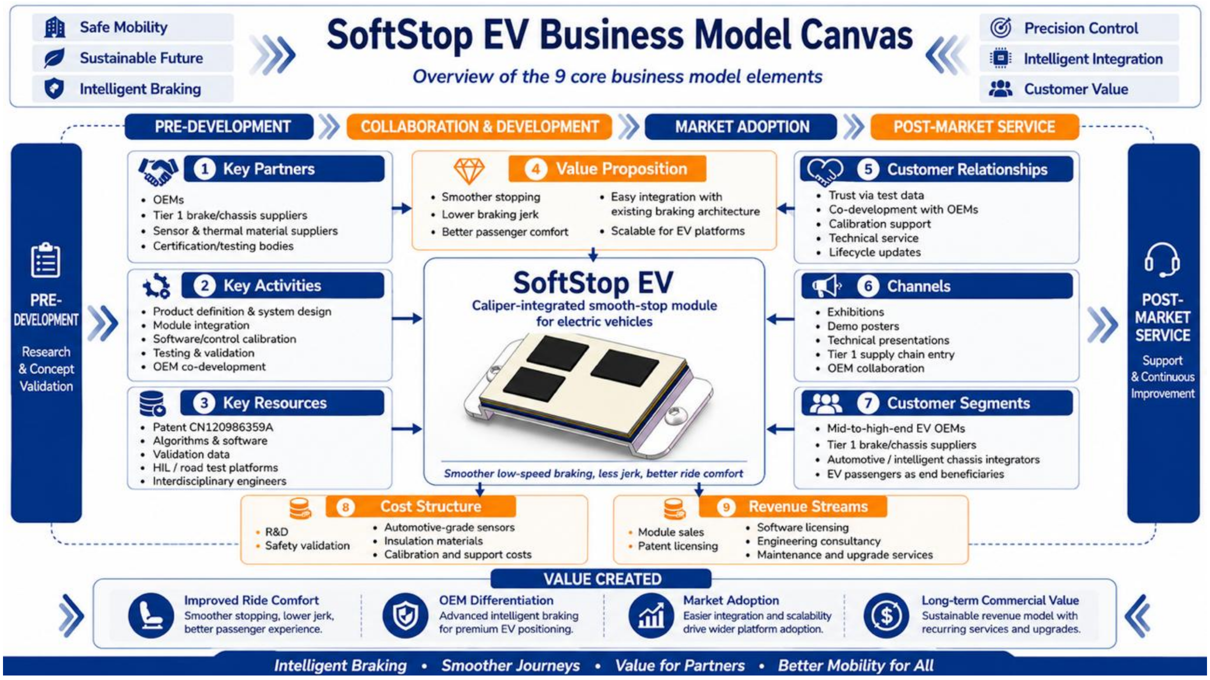

Fig. 1 SoftStop EV Business Model Canvas poster

As shown in Figure 1, by using the Business Model Canvas framework, this report analyses SoftStop EV from nine key aspects, including value proposition, customer segments, channels, customer relationships, revenue streams, key resources, key activities, key partnerships and cost structure. The analysis shows how the system can create value for OEMs, Tier 1 suppliers and end users, while supporting scalable adoption in the new energy vehicle supply chain [3][4]. The conceptual model of the front caliper finished product integrated with the SoftStop EV sensor system is shown in the Figure 2, and the final design shall be subject to the actual product specifications.

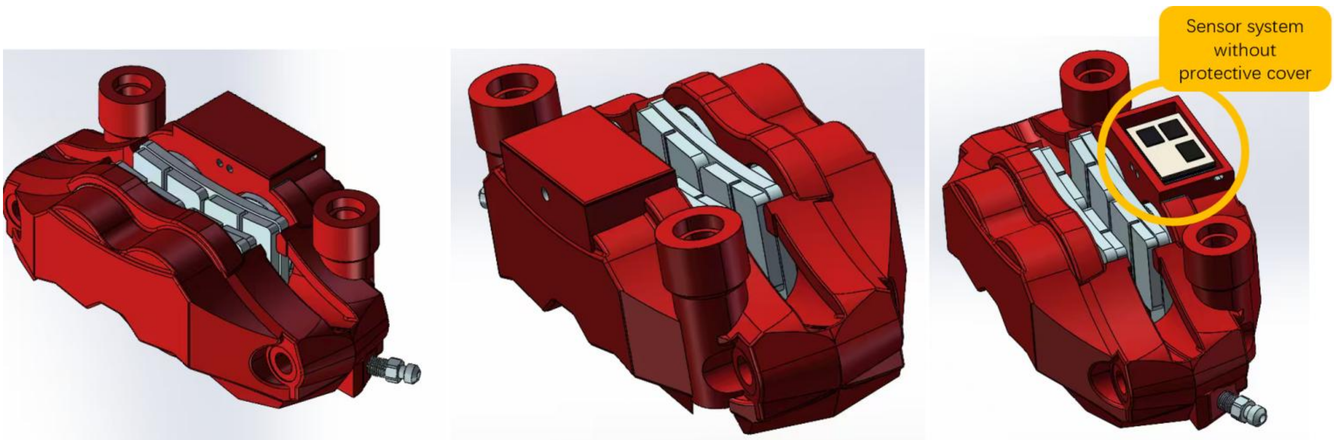

Fig. 2 Concept diagram of the product integrating the SoftStop EV sensor system

2 Value Proposition

The overall value proposition of SoftStop EV can be summarized in Figure 3, which connects its technical advantages, user experience benefits and commercial value.

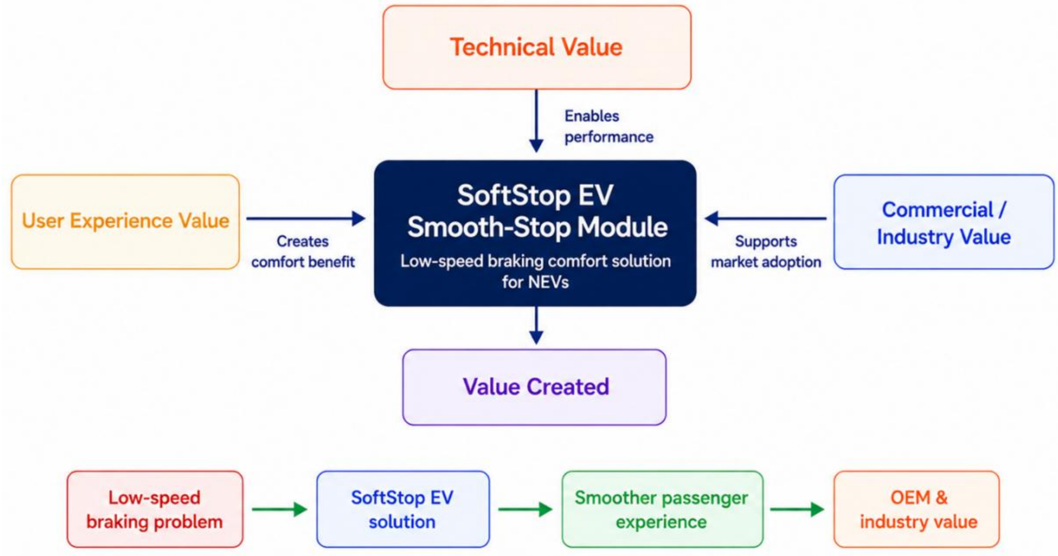

Fig. 3 Value proposition framework of SoftStop EV

2.1 Technical value/product advantage

The SoftStop EV System is the world's first automotive-grade integrated intelligent smooth-stop module specifically engineered for the front brake caliper operating conditions of new energy vehicles (NEVs). Based on the published invention patent held by the project team [1], and unlike industry-standard vehicle-level braking solutions, this product adopts a front caliper-integrated multi-sensor architecture. It integrates a Hall-effect speed sensor, a capacitive MEMS accelerometer, and a platinum resistance temperature compensation module at the foremost end of the braking actuation chain and incorporates a composite thermal insulation layer design to ensure stable module operation under extreme temperature conditions. This architecture reduces control latency to less than 10 ms and eliminates the need for modifications to the original vehicle ESP system architecture, significantly lowering adaptation and development costs for original equipment manufacturers (OEMs). To address sensor signal drift caused by high temperatures at the front caliper, the module integrates a patented temperature compensation algorithm combined with jerk closed-loop optimization technology, achieving an acceleration control accuracy of ± 0.001 m/s^2. The solution is specifically calibrated for the critical deceleration interval immediately before vehicle stop, while retaining full automotive-grade safety features including dual CAN bus redundancy, safety cut-off, and uphill compensation. It enables a smooth stop that is almost imperceptible to occupants without compromising braking safety, completely eliminating the pervasive "nodding judder" pain point during low-speed braking in conventional NEVs.

2.2 User experience/market value

SoftStop EV provides a significantly perceived ride comfort improvement for new energy vehicles, which has great value in terms of user experience. In low-speed stopping scenarios, such as traffic jams, vehicles often have setbacks and abrupt changes, which affect the user experience. Through the integrated multi-sensor, ECU control and temperature compensation module of the front brake caliper, the SoftStop EV can keep the acceleration smooth and reduce the jerk when the vehicle slows down, thus allowing passengers to feel a softer stop process. This function directly improves the user's riding experience in scenes such as urban roads, traffic lights and automatic parking [1][3].

For OEMs, SoftStop EV is not only a technical improvement, but also a reflection of market value. Smooth low-speed braking can be used as a differentiated selling point of high-end new energy models, improve brand image and product added value, enhance market competitiveness, and promote customer satisfaction and loyalty [3][4]. In addition, OEMs can visually present the advantages of SoftStop EV to potential customers through acceleration curves and visual experience demonstrations, strengthening technical credibility and market perception. This value reflects not only the perceived improvement of user experience, but also the competitive advantage of OEMs in the market, which fully highlights the comprehensive value of the product at the commercial level.

2.3 Commercial feasibility/industry value

At the industrial level, the value of SoftStop EV lies in its adaptability to original equipment manufacturers (OEM), Tier 1 suppliers and the wider new energy vehicle supply chain. The system is designed to work with existing ECU, CAN bus and front brake calliper architectures, which means that vehicle manufacturers would not need to completely redesign their braking systems. This can reduce the technical burden associated with development, integration and validation. For Tier 1 braking system suppliers, SoftStop EV could be integrated as an additional functional module within intelligent braking or brake-by-wire solutions, thereby strengthening their ability to provide more complete system-level offerings to OEM customers. This is important in the electric vehicle industry, where value creation increasingly depends on modular innovation, supplier collaboration and platform-based integration across the automotive value chain [5][6]. In addition, the technology has strong potential for multi-model platform applications, as it can be adapted to different vehicle weights, braking characteristics and chassis structures across entry-level, mid-range and premium new energy vehicles. The system can also provide measurable evidence of comfort improvement through acceleration curves, jerk indices and low-speed stopping stability data. These quantifiable indicators allow OEMs to demonstrate product advantages more convincingly and support differentiation in an increasingly competitive electric vehicle market. Therefore, SoftStop EV is not only a technical innovation, but also a commercially feasible solution with potential for scalable adoption across the new energy vehicle supply chain.

3 Customer Segments

Our project provides an intelligent SoftStop sensor system intended to be used in the front brake calipers of new energy vehicles. This product is not sold as an aftermarket retail product to all car owners. On the contrary, it is a complete module, which should be closely related to the electronic control unit of the vehicle as well as its base braking architecture. Therefore, our business model is mainly B2B-oriented rather than retail sales. Our main clients are the mid-to-high end new energy vehicle manufacturers, who are our direct sales targets and technology partners. The existing environment of the electric vehicle industry is very competitive and the fundamental technology of batteries and motors is already available in most of the brands. Therefore, smart chassis tuning and ride comfort are the new important selling points, which allow premium car companies to develop a distinct market position.

The secondary clients in our business are the Tier 1 brake system vendors and automotive electrical integrators such as Bosch and Continental. The sensor package is considered a sub-system, which is sold to these Tier 1 firms. The Tier 1 companies assist us to incorporate our product into the approved supply chain of the large car brands. Moreover, the end-users of our product are ordinary motorists and passengers who are very susceptible to motion sickness and unpleasant braking but do not purchase our module on their own. Therefore, their demand for a comfortable ride drives the car manufacturers to adopt our product. The following figure presents the hierarchical structure of the target market.

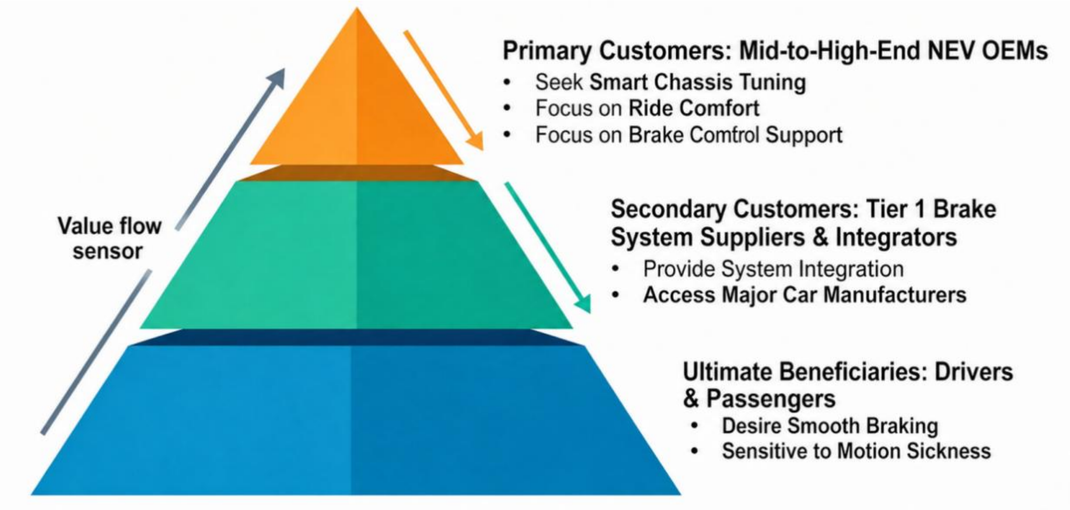

Fig. 4 The Three-Tier Customer Segmentation Pyramid for the SoftStop System

We have a logical customer segment strategy which fits well with the actual business and engineering challenges faced by automobile producers: premium car brands require a smooth low-speed ride; car manufacturers demand high-end integration and safety without changing existing hydraulic brake safety levels [7], and they need to address the problem of sensor overheating and signal loss. These challenges they encountered can serve as our unique selling propositions, which is why we target these customer tiers. The integration process in the automotive supply chain is shown in the value flow diagram.

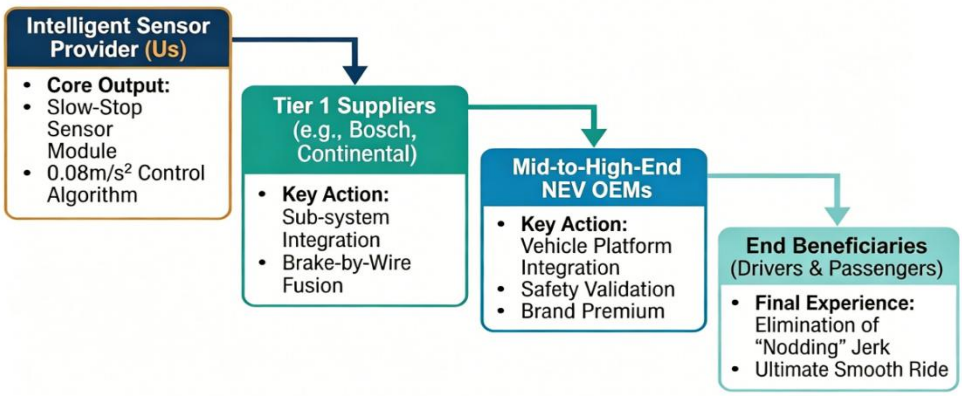

Fig. 5 B2B Integration Path and Value Flow from Sensors to End Beneficiaries

Eventually, our module provides car manufacturers with a chance to demand a higher selling price. By addressing the problem of low-speed stopping directly, the premium quality of the car is increased. The whole automotive industry is shifting towards intelligent and more efficient braking systems [8]. Therefore, our customers are more willing to spend money on chassis modules which can provide a significantly improved driving experience.

4 Channels

SoftStop EV is an engineering product that integrates with the vehicle braking system, ECU control logic, and electronic architecture. It uses multiple sensors and a temperature compensation module to monitor low-speed braking. This improves stopping smoothness and passenger comfort [1]. To enter the market, SoftStop EV is best introduced via B2B channels, starting with auto parts suppliers and then gradually integrated into vehicle mass production.

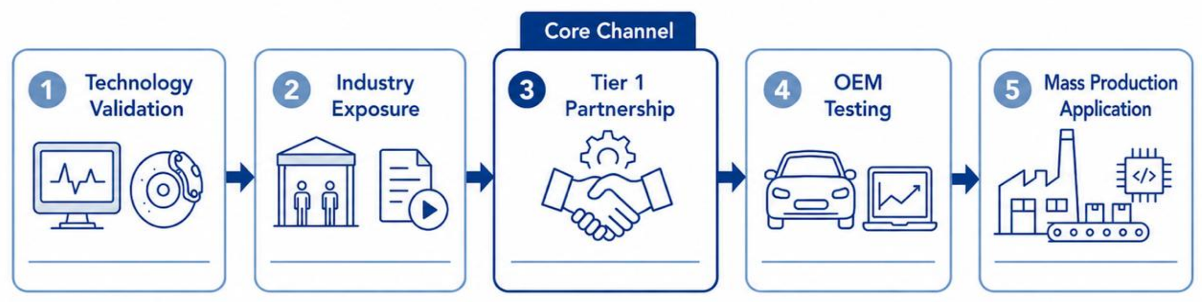

Fig. 6 Market entry lifecycle of SoftStop EV

Figure 6 shows the market entry of SoftStop EV in a series of steps, beginning with technology validation and followed by supply chain introduction. The automotive supply chain channels fall into three categories (Table 1). Tier 1 partnerships serve as the core channel, OEM cooperation is a longer-term objective, and exhibitions help attract early customer attention [2] [9].

Table 1 SoftStop EV's main channels, target customers, and core functions

Channel	Target	Priority	Function
Tier 1 Partnership	Brake / chassis suppliers	High	Enter mass production supply chains
OEM Partnership	EV manufacturers	Medium-high	Gain testing and vehicle nomination opportunities
Exhibition & Materials	Trade shows, white papers, demo videos	Medium	Show value and attract customers

Working with a Tier 1 braking system or smart chassis supplier is the channel with the highest priority to advance. The supply chain system in the automotive industry is relatively strict, and new products usually need to go through a long verification cycle if they want to directly enter the mass production platform of the OEM. In contrast, established Tier 1 suppliers already have customer resources for brake calipers, line controls, and chassis control systems, and are more familiar with mass production manufacturing processes. Research on EV supply chain points out that supplier integration helps to improve the supply chain competitiveness of OEMs [9].

SoftStop EV can enhance the user experience for OEMs, potentially increasing vehicle sales. The linear control system is a key component of the intelligent chassis and affects both braking efficiency and ride comfort [3]. SoftStop EV can target high-end EV models, such as Tesla and BYD, and demonstrate its value through acceleration and jerk comparisons. Figures 7 and 8 show the examples of professional comparison curves for acceleration and jerk, respectively, and a smoother curve means a more comfortable ride experience.

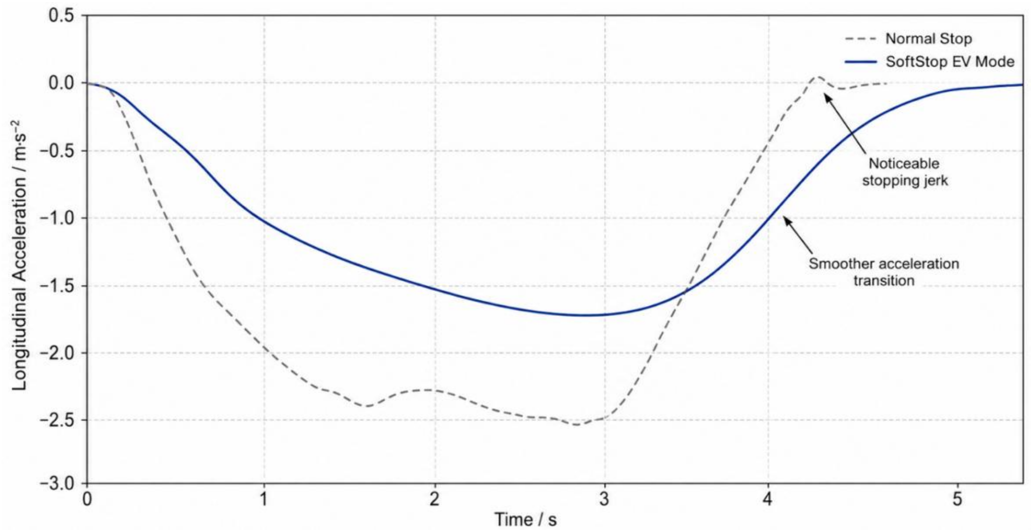

Fig. 7 Acceleration comparison curve

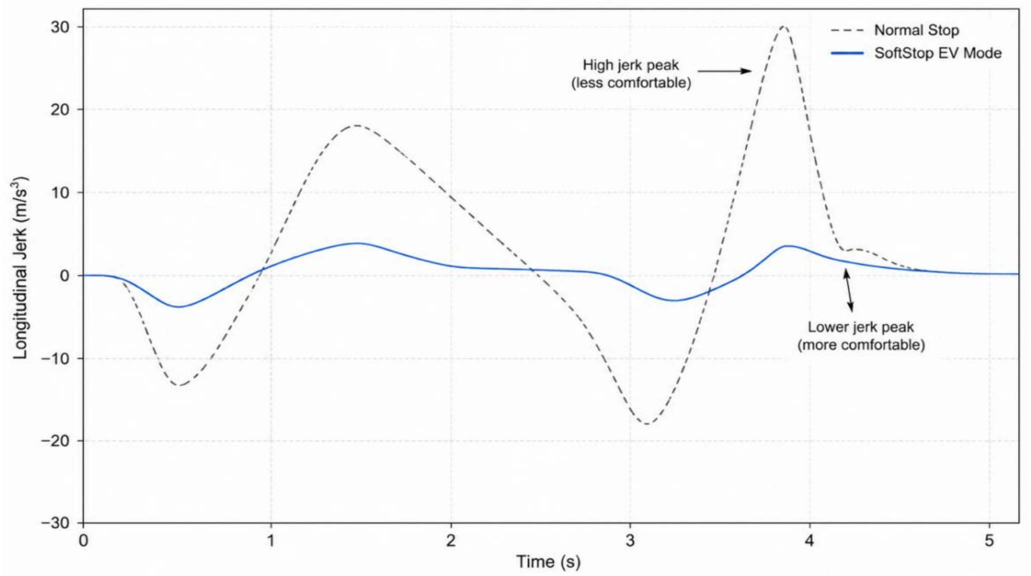

Fig. 8 Jerk comparison curve

While trade shows and technical presentations are critical to early customer acquisition, it is difficult to visually and clearly present the effects of SoftStop EV to customers using text alone. Therefore, the team can visualize the product function by comparing the schematic diagram to show the ride comfort and ride comfort of low-speed braking. Figure 9 shows the poster case. The smooth deceleration and comfort improvement of SoftStop EV compared with ordinary braking are intuitively presented through the car diagram, simplified acceleration comparison curve and customer experience icon, conveying the core value: smoother stopping, less stuttering and improved user experience of high-end EV [4]

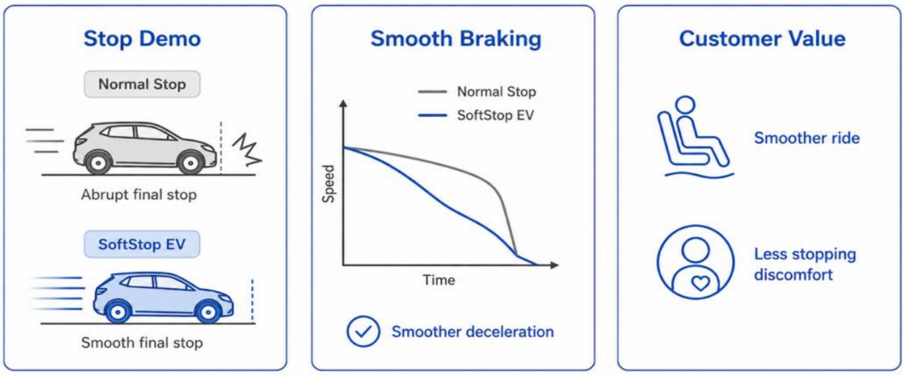

Fig. 9 SoftStop EV exhibition demonstration poster

Overall, SoftStop EV's channel strategy can be summarized as an early stage, where the team should mainly attract attention through industry shows; In the middle stage, we should rely on Tier 1 cooperation to promote mass production introduction. In the later stage, the scope of application can be expanded by platform algorithm authorization.

5 Customer Relationships

SoftStop EV's customer relationship is a long-term engineering partnership. The system collects the low-speed braking state through the multi-sensor and temperature compensation module, and the front brake caliper pressure is adjusted by the ECU to reduce the frustration of the vehicle when it stops [1]. Therefore, the relationship between SoftStop EVs and customers should shift from "selling products" to "jointly implementing vehicle-level functions". The customer relationship can be summarized in three phases, as shown in Figure 10.

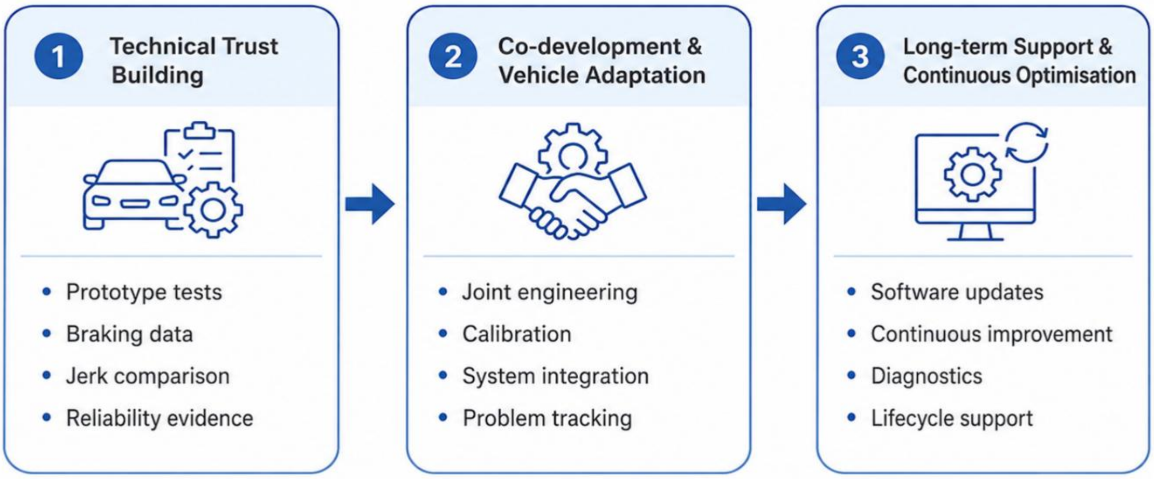

Fig. 10 Customer relationship lifecycle of SoftStop EV

The first phase is to build technical trust. Brake related products are highly safety sensitive and Tier 1 suppliers will not adopt new systems based on concept alone. Therefore, the team needs to prove the engineering feasibility with procedures such as prototype testing, acceleration curves, and jerk comparison. Research on B2B value co-creation points out that enterprise customer relationship is usually not a one-way transaction, but a process of value co-creation between suppliers and customers [10]. Thus, early relationships focus on building trust with reliable data.

The second phase is joint development and vehicle adaptation. Industry research shows that OEMs and Tier 1 suppliers in the automotive industry usually need deep collaboration in new function development and system integration [2]. The performance of SoftStop EV on different models is affected by vehicle weight, suspension structure, braking system, etc. Therefore, system integration and parameter calibration should be carried out for specific models. Table 2 summarizes how the SoftStop EV should manage the relationship with different customer groups.

Table 2 SoftStop EV customer relationship focus by customer group

Customer group	Relationship focus	Main support
Tier 1 suppliers	System integration	Interface documents, module adaptation, production support
OEMs	User-experience validation	Vehicle testing, calibration, comfort evaluation
Mass-production customers	Continuous optimisation	Software updates, diagnostics, quality tracking

In the final phase, SoftStop EV provides long-term support after mass deployment. As vehicles move toward software-defined architectures, functions increasingly rely on updates and lifecycle management. SoftStop EV can offer software updates and optimize control parameters over time to maintain consistent performance and comfort for customers [11].

The team should also implement a clear customer support process. Studies on B2B after-sales management indicate that complex products require ongoing technical support and dedicated attention to key customers to maintain long-term partnerships [12]. SoftStop EV can assign key account managers and an engineering support team to regularly provide test reports, software updates, and quality feedback.

In general, the customer relationship of SoftStop EV should focus on "trust, co-creation and continuous service". Trust is built with test data in the early stage, vehicle adaptation is completed through joint development in the middle stage, and long-term cooperation is maintained through software upgrades and technical support in the later stage.

6 Revenue Streams

The revenue model for the proposed SoftStop sensor system integrated into the front brake callipers of new energy vehicles is designed as a diversified business-to-business model. As an automotive engineering technology, the system should not rely only on one-off hardware sales. Instead, revenue can be generated through hardware module sales, patent licensing, software and algorithm services, vehicle calibration, after-sales maintenance, system upgrades, and joint development projects. This approach is consistent with the Business Model Canvas, which explains that revenue streams should show how a company captures value from the products and services it offers to customers [13].

The first and most direct revenue stream is the sale of physical sensor modules to new energy vehicle manufacturers, braking system suppliers, and intelligent chassis companies. The system includes speed sensors, acceleration sensors, temperature compensation, signal processing units, CAN bus communication, ECU control logic, and pressure regulation functions. These elements help reduce low-speed braking “nodding” and improve passenger comfort during the final stopping stage [1]. The company could sell the system on a per-vehicle basis, with each vehicle using two front-calliper sensor units. Once adopted by an original equipment manufacturer, large-scale production could create stable and scalable income.

A second revenue stream is patent and technology licensing. The patent describes a sensor system integrated into the front brake callipers of new energy vehicles, including control logic, safety cut-off, uphill compensation, CAN bus communication, and temperature compensation [1]. These technologies could be licensed to automotive manufacturers or Tier 1 suppliers through a one-time technology transfer fee or royalties based on vehicle production volume. This model allows the company to monetise its intellectual property without taking on the full cost and risk of mass production. It also reflects the idea that firms can gain value not only from products, but also from the way technology and intellectual property are commercialised [14].

Thirdly, the company can generate income from software algorithms and control strategy services. The value of the system depends not only on hardware, but also on real-time data processing and ECU-based braking control. Since different vehicle models have different weights, braking characteristics, regenerative braking strategies, and chassis structures, each platform will require specific calibration and validation. Therefore, the company can charge for algorithm licensing, vehicle-specific calibration, testing support, and annual software updates. This is particularly important for safety-related automotive systems, which must be carefully developed and maintained to meet functional safety expectations [15].

Another important revenue source is after-sales service and system upgrades. Because the product is related to braking performance and safety, customers will require diagnostics, maintenance, calibration, and software updates. The company could provide technical support, remote upgrades, data analysis, and diagnostic tools to vehicle manufacturers, service centres, and fleet operators. This is especially valuable for taxis, ride-hailing vehicles, logistics vehicles, and autonomous fleets, where frequent stopping increases the importance of braking comfort and system durability.

Finally, during the early commercialisation stage, the company can earn revenue through joint research and development projects, prototype testing, and engineering consultancy. Cooperation with car manufacturers, universities, testing centres, and braking system suppliers can provide early cash flow while helping improve product reliability. Overall, the proposed revenue streams combine hardware sales, licensing, software services, calibration, maintenance, upgrades, and consultancy, supporting both short-term market entry and long-term profitability.

7 Key Resources

The key resources of this project can be divided into hard and soft resources. Categorized into intellectual, physical, human, financial & supply chain per the Business Model Canvas, they form the core foundation for delivering the "smooth-stop" sensor system to original equipment manufacturers (OEMs) (Fig. 11).

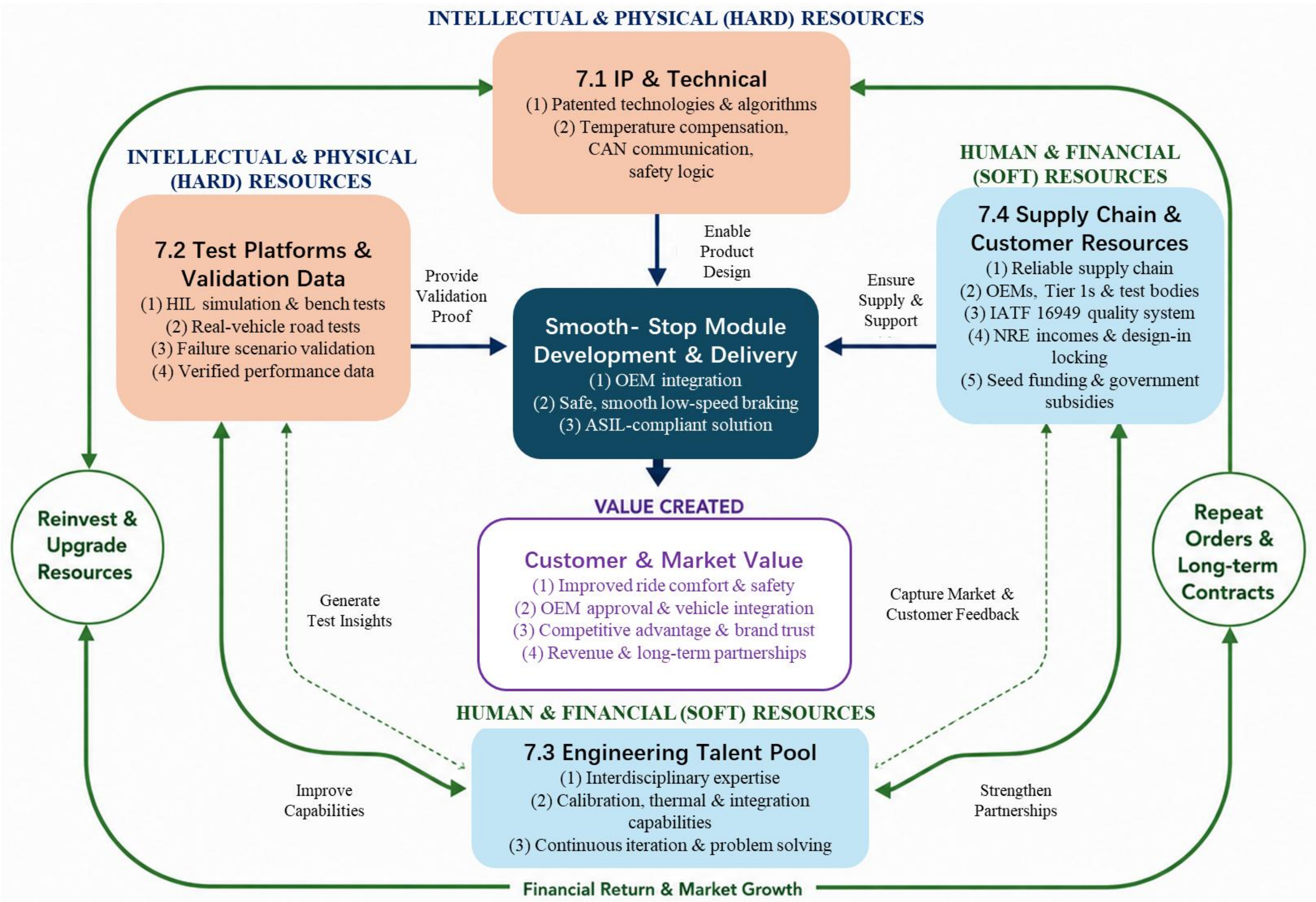

Fig. 11 Overview of key resources value creation and closed-loop logic

7.1 Intellectual Property and Technical Resources

The most critical resource of this project is the granted technical solution and control logic of the "smooth-stop" system (Patent Publication No.: CN 120986359 A) [1], which covers all functional modules including speed signal acquisition, acceleration monitoring, temperature compensation, CAN communication and safety cut-off, as well as the control logic and protection strategies.

In the B2B automotive supply chain, patents are the lifeline for maintaining pricing power. This IP prevents reverse engineering by major players like Bosch and Continental. Our team masters all technical details from mechanical design to control algorithms, resolving OEM integration issues rapidly. Going forward, we will continue to apply for utility model patents and software copyrights based on real-vehicle test data.

7.2 Test Platforms and Validation Data

Prototypes and test platforms are also critical hard resources, including hardware-in-the-loop (HIL) simulation platforms, precision thermal characteristic laboratories, and cooperative real-vehicle test sites [16]. Since the product needs to operate stably in environments with high and low temperatures, vibration shock and water immersion, it must rely on bench testing, road testing and failure scenario validation to optimize the design.

Validated data itself is a valuable asset and serves as the indispensable physical endorsement for the product to obtain OEM technical approval and automotive-grade certification (ASIL standard).

Currently the development of 3 functional verification prototypes is planned, and thermal characteristic testing will be conducted simultaneously using laboratory resources from local universities. In the future, we will apply for test equipment subsidies from the Ministry of Science and Technology of China and share real-vehicle test sites with OEMs.

7.3 Interdisciplinary Engineering Talent Pool

Interdisciplinary expertise is our most valuable soft resource. The system requires expertise in mechanical design, automotive electronics, embedded software and brake integration. Experience in sensor placement, brake calibration, thermal management and automotive-grade communication, in particular, is far more valuable than hardware procurement for B2B automotive-grade products.

Future engineering graduates responsible for structural optimization, thermal insulation material iteration, CAN bus communication and encryption algorithms form the core of the team. Specialized human resources are the only way to deliver the complex custom calibration required for B2B integration [17]; simultaneously, by cultivating this high-value talent pool, this project directly contributes to the transformation of the UK/CHN economy towards advanced manufacturing. Going forward, we will recruit several additional interdisciplinary engineers to provide full-process technical support to OEMs.

7.4 Supply Chain and Customer Resources

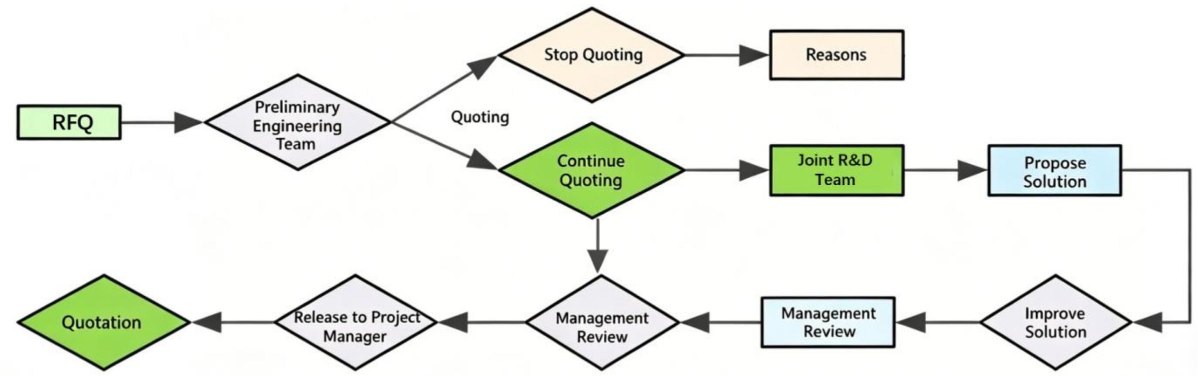

Fig. 12 Quotation process for automotive electronic components

The product requires continuous supply of sub-sensors, thermal insulation base materials, wiring components and original brake systems, and relies on in-depth engineering collaboration based on the IATF 16949 quality management system. Meanwhile, cooperative relationships with OEMs, component suppliers and testing and certification bodies also directly impact the project's introduction and commercialization progress. We have planned to submit a Request For Quotation (RFQ) to leading Tier 1 brake system suppliers (Fig. 12) and subsequently sign a strategic cooperation agreement.

Furthermore, the NRE (Non-Recurring Engineering) fees collected from OEMs during the joint R&D phase serve as our crucial financial resource in the early-stage. Based on the typical "pre-installation lock-in" characteristic of the automotive industry, it creates high OEM switching costs post-validation.[18]. This core "design-in" market resource ensures stable revenue streams throughout the vehicle's lifecycle.

8 Key Activities

To successfully translate the "smooth-stop" sensor system into a commercially viable module that can be directly adopted by original equipment manufacturers (OEMs), the key activities of the product span the entire lifecycle from research and development (R&D) to mass production [19]. Based on the commercial essence of this product as a "B2B pre-installed integrated solution", OEMs are purchasing not merely a standalone sensor, but a complete "comfortable braking solution".

8.1 System Definition and Functional Requirements Confirmation

The technical solution shall be translated into a vehicle-level product definition, with clear specifications of low-speed smooth stop objectives, trigger conditions, safety cut-off conditions and other critical parameters. While the patent has explicitly defined core elements including the low-speed threshold, acceleration magnitude and temperature compensation mechanism, this product is a configurable pre-installed module. Clear definition of functional boundaries and specific braking strategies establishes the core for subsequent R&D efforts [20].

Therefore, system requirements definition serves as the fundamental starting point. This phase is expected to reduce the amount of rework by more than 30%. Full vehicle ECUs would not be produced in the product, focusing solely on low-speed smooth stop optimization.

8.2 Joint R&D and NRE Customization

Collaborate with OEMs on NRE (One-Time Engineering Cost) projects. The Hall-effect speed sensors, MEMS accelerometers and composite thermal insulation brackets should be precisely integrated into the braking systems of target vehicle models, hence, it requires the development of a vehicle-ready integrated module.

The engineering focus of this product lies not in individual components, but in the overall system integration reliability, as evidenced through the structural design in the patent. A more detailed operational principle is illustrated (Fig. 13). This ensures seamless OEM integration. Meanwhile, the basic sensors will be sourced from certified automotive suppliers rather than manufactured in-house.

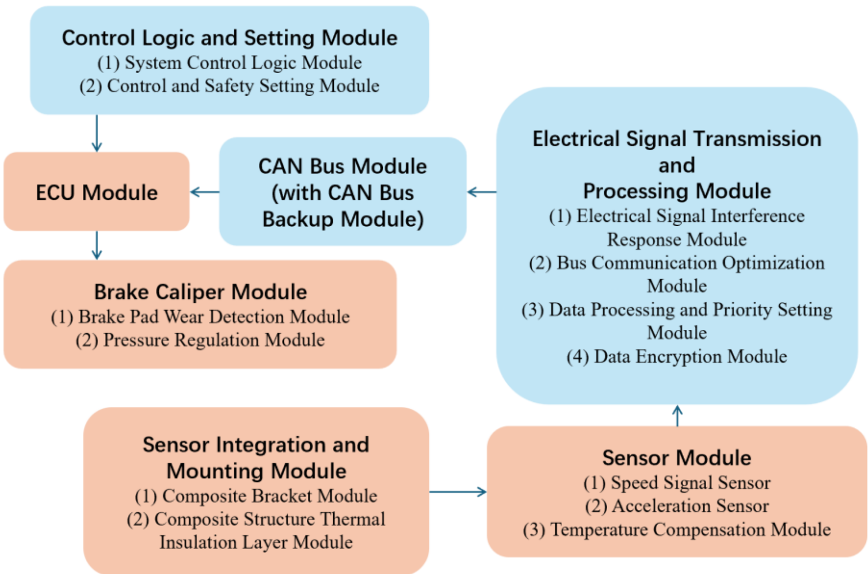

Fig. 13 Integrated 'smooth-stop' sensor system architecture and operation flow

8.3 Software Development and Protocol Calibration

The product's differentiated value lies, more importantly, in achieving precision and stability under software control through hardware-software integration. Program development is conducted following pre-modelling of the control system, incorporating configurations for low-speed braking control, temperature compensation, signal correction, CAN communication, data prioritization, dual redundant channels and uphill compensation.

The control logic must enable smooth vehicle braking during the final deceleration phase while maintaining robust safety fallback mechanisms. The specific braking solution is also determined by the customer, enabling product customization. Precision calibration and algorithm integration constitute the core commercial advantage of this product.

8.4 Test Verification and Mass Production Implementation

Test verification is designed to thoroughly eliminate OEMs' safety concerns regarding the introduction of new hardware, thereby translating patented technology into commercially viable mass-production modules [21]. Only through comprehensive bench testing, road testing and failure scenario validation (e.g., the protective efficacy of the composite thermal insulation layer for sensors at temperatures above 300°C, system robustness under complex road conditions) will OEMs recognize it as an adoptable pre-installed module. The prototype vehicle platform has been fully established (Fig. 14), with the same control algorithm applied to the model vehicle, verifying the feasibility of the braking scheme.

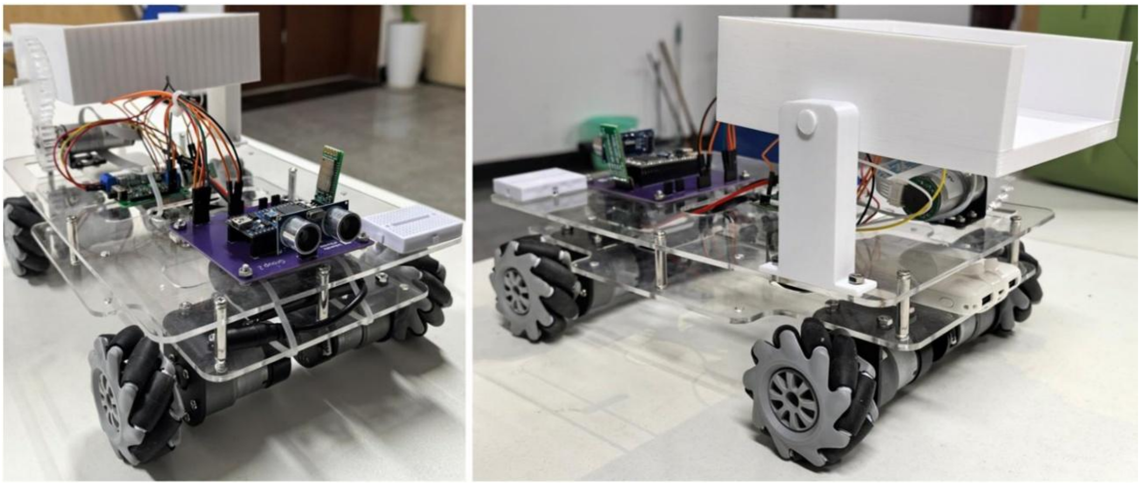

Fig. 14 A model vehicle implementing softstop using the same control algorithm.

Upon completion of validation, supply chain collaboration, process freezing, quality documentation compilation and customer technical alignment are required. Thereafter, the project can smoothly transition from the single-component phase to the vehicle platform integration phase. All physical modules (in red) and communication links (in blue) that require test verification have been listed in Figure 13. The three most critical sub-sensors are highlighted in the SoftStop EV sensor modeling conceptual diagram (Fig. 15).

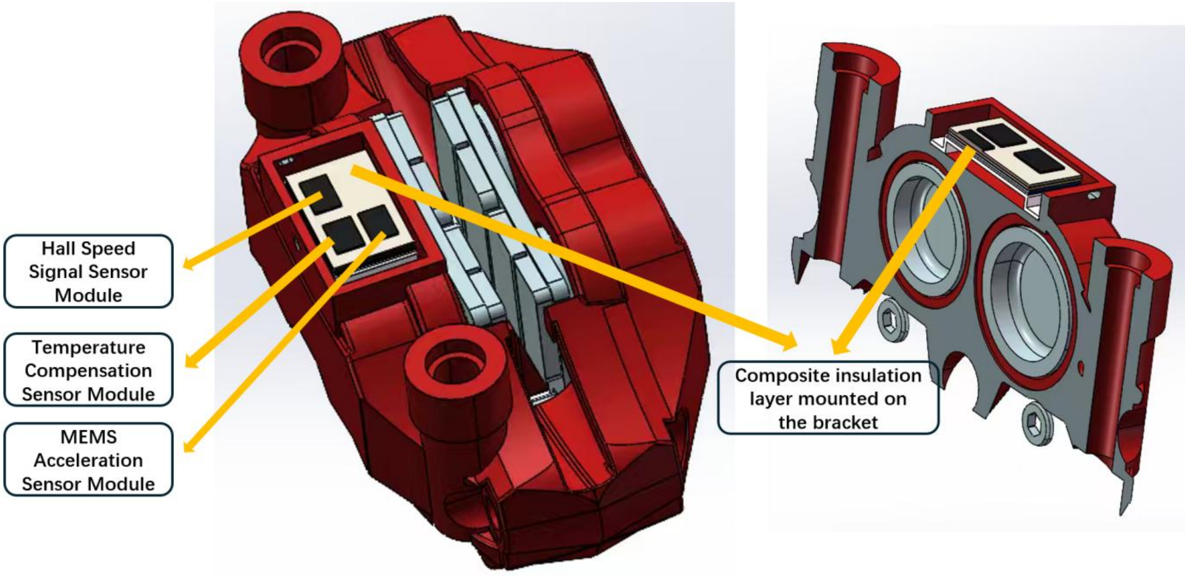

Fig. 25 The three sensor modules and their corresponding thermal insulation layer positions illustrated in the SoftStop EV sensor modeling conceptual diagram

9 Key Partnerships

Commercializing a "Gentle Stop" module—specifically designed to stop the longitudinal "nodding" effect during NEV deceleration—isn't just about selling a part. Because this module is integrated directly into the front brake calipers and the vehicle's electronic brain, we have to build a B2B network that can handle the heavy demands of the automotive supply chain.

9.1 The Anchors: Original Equipment Manufacturers (OEMs)

Automakers like Volkswagen or Toyota are our primary customers and strategic anchors. We can't sell this as a simple add-on; it has to be part of the car's design from the start. A "joint development" model is the only way this works. We need the OEM to provide technical specs and testing environments, while we provide them with a way to differentiate their brand through superior ride quality. This partnership also allows for platform-level scaling, where our braking logic can be refined over time via over-the-air (OTA) updates [22].

9.2 The Foundation: Tier 1 Braking Partners

We need to work closely with the giants of the braking world, such as Robert Bosch GmbH and Continental AG. They provide the "One-Box" hardware that our system plugs into, like Bosch's Integrated Power Brake or Continental’s MK C2 [23], shown as Figure 16. These partners are vital because they handle the baseline safety layers like ABS and ESP. Our module acts as a high-precision "fine-tuner" for the final meter of the stop, but it must blend perfectly with their regenerative braking logic to keep the pedal feel consistent for the driver [23].

Fig. 36 Bosch's Integrated Power Brake

9.3 Specialty Tech: Sensors and Materials

The "intelligence" of the module depends on specialized providers:

Semiconductor Partners: We rely on companies like STMicroelectronics for high-precision motion sensing. Their AEC-Q100 qualified MEMS chips can detect tiny acceleration shifts that are invisible to the human eye, ensuring the system reacts in milliseconds [24].
Material Partners: Since the sensors sit right on the brake caliper, heat is a massive threat—calipers can easily hit 300°C. We need partners like Aspen Aerogels to provide PyroThin insulation barriers. These materials have ultra-low thermal conductivity, which is the only thing keeping our electronics from frying in that harsh environment [25].

9.4 The Gatekeepers: Certification Bodies

In a safety-critical industry, we can't move forward without the stamp of approval from bodies like TÜV SÜD. They act as "gatekeepers" who validate that our module meets the ISO 26262 ASIL D standard—the highest level of functional safety [26]. Without this certification, no OEM would risk putting our module into a mass-produced vehicle. The V-Diagram is shown as Figure 17.

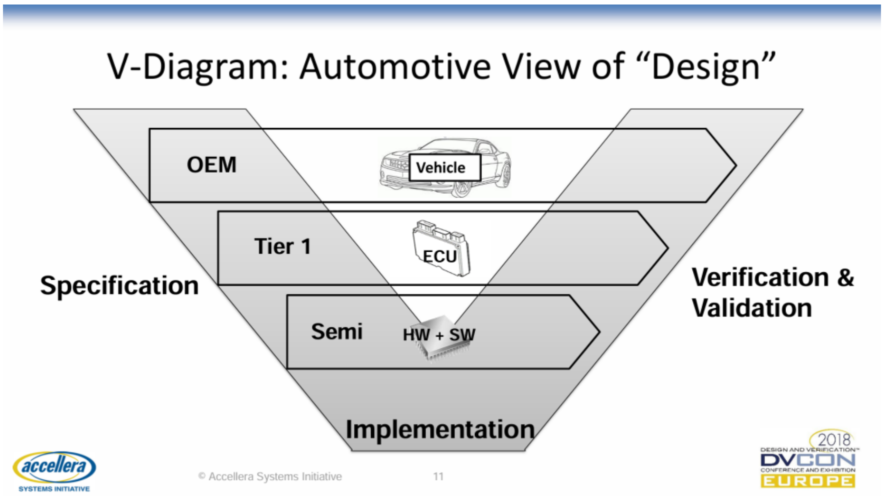

Fig. 17 Automotive Collaboration V-Model illustrating the roles of OEM, Tier 1, and Semiconductor partners in the integrated design and validation process

The main reason for this web of partners is to share the risk and lower the barrier to entry. Developing safety-critical automotive tech from scratch is too expensive. By using modular platforms and shared-part strategies with these established names, we can cut production costs by an estimated 15-20% and avoid the typical R&D pitfalls of working alone [27].

10 Cost Structure

The cost structure of our project exhibits a typical automotive hardware profile, characterized by high initial fixed costs followed by decreasing variable costs driven by economies of scale. Designed specifically for mid-to-high-end new energy vehicles, expenditures strictly center on core system development, safety redundancy, and physical integration. Table 3 categorizes the core cost drivers of the SoftStop EV module to map out the corresponding strategic rationale.

Table 3 Cost Structure and Strategic Rationale Breakdown for the SoftStop EV Module

Cost Category	Key Cost Items	Example Components / Activities	Target Suppliers / Partners
Fixed Costs	Control Logic & Algorithms	Hall & MEMS sensor synergistic software, AES-128 encryption	Internal R&D Team
	Functional Safety Certification	ISO 26262 ASIL validation, environmental testing	SGS, TÜV SÜD
Variable Costs	Automotive-Grade Sensors	Capacitive MEMS accelerometers, Hall speed sensors	Infineon, NXP
	Thermal Insulation Materials	Nano-aerogel (≤0.013W/(mK)), ceramic fiber pads	Cabot Corporation, Unifrax
Operational Costs	System Calibration & Integration	Platform-specific ECU matching, chassis tuning	Joint Development with OEMs (e.g., NIO, BMW)

10.1 Fixed Costs for Initial R&D and Redundancy Design

The largest initial expense involves developing the multi-sensor synergistic control algorithm and designing data security protocols. Massive capital must be invested to develop the underlying logic for

Hall speed sensors and capacitive MEMS accelerometers [1]. This allows for precise caliper force control during low-speed vehicle stops.

This substantial research investment is crucial for building our technical moat. A dual-sensor redundancy backup system was specifically designed. This addresses potential occasional failures of high-precision MEMS sensors caused by bumps, vibrations, or temperature fluctuations. An AES-128 symmetric encryption algorithm was also developed within the transmission module to prevent hackers from tampering with vehicle bus data. While these rigorous designs increase early-stage costs, they are necessary prerequisites to eliminate automakers' safety concerns. Securing ISO 26262 ASIL certification from bodies like SGS is a costly but mandatory threshold for automotive supply chain entry [28].

10.2 Variable Costs for Hardware Materials and Physical Integration

Variable costs primarily involve procuring automotive-grade chips (e.g., from Infineon or NXP) and specialized integration materials. A three-layer composite thermal insulation structure is adopted to withstand front brake caliper environments exceeding 300°C. This structure comprises nano-aerogel (≤0.013 W/(mK), e.g., from Cabot Corporation [29]), ceramic fiber pads (e.g., Unifrax), and mica sheets.

While these specialized polymers increase the single-unit bill of materials cost, they are physically indispensable. Relying on these expensive materials allows the sensor surface temperature to be strictly controlled below 120°C when integrating with premium calipers like Brembo. This uncompromising protection guarantees measurement accuracy under extreme heat, ultimately fulfilling the comfort promise of delivering smooth stops—the exact value automakers seek to enhance vehicle premiumness.

10.3 Operational Costs for Management and System Maintenance

As a technology supplier, integration development fees are incurred based on specific projects. Long-term bus communication maintenance and specialized system calibration costs are also generated.

This SoftStop system requires constant high-frequency data interactions with the vehicle's electronic control unit. However, initial development expenses are rapidly diluted through modular designs and standardized interfaces as the technology expands into mass production across premium platforms (e.g., BMW, NIO). The economies of scale inherently brought by mass production sharply reduce unit purchasing prices for sensors and marginal manufacturing costs, guaranteeing ample long-term profitability while providing end-users with an ultimately smooth experience.

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